9.9 The Birthday Problem

WORKSHEET

The question

STEP II 2007, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q13

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Why is it easier to calculate the probability that *no* two guests share a birthday? Line the guests up: what is the chance that the second avoids the first's birthday? That the third avoids both?

STAGE 1 A useful approximation

Ask yourself What is the tangent line to $y = e^{-x}$ at $x = 0$?

STAGE 2 The probability of a shared birthday

Ask yourself Look at the exponent: $\frac{k(k-1)}{730} = \binom{k}{2} \Big/ 365$. What does that say in words?

STAGE 4 Sharing with the host

Ask yourself For each guest, what is the probability of *not* sharing the host's birthday? Are the guests independent of each other in this respect?

Try this next

1 **The first repeat.** (From STEP II, 2014.) A random number generator produces integers from 1 to n, independently and equally likely. Let X be the position of the first repeat. Find $\mathrm{P}(X = r)$ for $2 \le r \le n+1$, and deduce that

$$\frac{1}{n} + \left(1 - \frac{1}{n}\right)\frac{2}{n} + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\frac{3}{n} + \cdots = 1.$$

Show that $\mathrm{E}(Y) = \sum_{k=1}^{N} \mathrm{P}(Y \ge k)$ for any random variable taking values $1, \ldots, N$, and hence that

$$\left(1 - \frac{1^2}{n}\right) + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2^2}{n}\right) + \left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\left(1 - \frac{3^2}{n}\right) + \cdots = 0.$$

2 **Three in common.** Roughly how many people are needed for a better-than-even chance that some *three* share a birthday? (*Hint:* the expected number of triples sharing a birthday is $\binom{k}{3}/365^2$. Set this to about $\ln 2$.) The true answer is 88.

3 **Near misses.** How many people are needed for an even chance that two have birthdays on the same day or on adjacent days?

4 **Uneven birthdays.** Real birthdays are not uniform. Show that if the days have probabilities $p_1, \ldots, p_{365}$, the chance that two people share a birthday is $\sum p_i^2$, and that this is smallest when all p_i are equal. (So non-uniformity only makes matches more likely.)

Further reading

- **William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1** (Wiley), Chapter II. The birthday problem and its approximations.
- **Persi Diaconis & Frederick Mosteller, "Methods for studying coincidences"**, *Journal of the American Statistical Association* 84 (1989). Why coincidences are common; very readable.
- **David Hand, *The Improbability Principle*** (Bantam). The square-root law and other reasons why unlikely things happen all the time.

The full guided solution, and what the question leads to at university, is in Section 9.9:
https://beyondstep.keithmcnulty.org/ch09/birthday.html